BLOOD DONATION ANALYTICS DASHBOARD

Project Documentation & Technical Reference

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10,000 Donor Records | 15 States | 20 Cities | 10 Donation Centers

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1. Executive Summary

This document provides a comprehensive technical and analytical reference for the Blood Donation Analytics Dashboard, built on a dataset of 10,000 donor records spanning India. The dashboard is designed to support decision-making for blood banks, healthcare administrators, and public health organizations by surfacing trends in donor demographics, eligibility, geographic distribution, and donation centre performance.

The dataset captures donor registrations from February 2008 through November 2025, making it a rich longitudinal resource for understanding donation behaviour over time. The Excel workbook is structured with six sheets: a raw data layer, multiple pivot-based analytical tables, and dedicated dashboard and demographic/geographic views.

2. Project Overview

2.1 Purpose & Objectives

The primary goal of this project is to convert raw blood donation registration data into actionable insights that can:

- Identify the most and least common blood groups in the donor pool
- Measure donor eligibility rates and flag key disqualifying conditions
- Analyse geographic spread of donors across states and cities
- Benchmark donation centre performance by volume and average donations per donor
- Understand demographic patterns (age, gender) driving donation frequency

2.2 Scope

The project covers all registered donors in the dataset. Country scope is restricted to India. There are 15 states, 20 cities, and 10 donation centres represented. The time horizon spans from 2008 to 2025.

3. Workbook Structure

The Excel workbook contains six sheets, each serving a distinct purpose in the analytics pipeline:

Sheet Name	Type	Description
About	Information	Overview/metadata sheet (design layer, currently no raw data)
Dashboard	Visual	Main interactive dashboard with charts and slicers
Demographics	Analytics	Demographic breakdown (age, gender, eligibility analysis)
Geographics	Analytics	State and city-level geographic distribution visuals
Table1 (2)	Pivot Data	Source pivot tables feeding dashboard charts

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Sheet Name	Type	Description
Data	Raw Data	Master dataset — 10,000 rows × 18 columns

4. Data Dictionary

The primary data sheet ('Data') contains 18 columns. Each column is described below:

Field Name	Data Type	Nullable	Description
Donor_ID	String	No	Unique identifier for each donor (format: DNR000001 … DNR010000)
Full_Name	String	No	Full name of the donor
Gender	Categorical	No	Female / Male / Other
Age	Integer	No	Donor age in years (18–65)
Blood_Group	Categorical	No	ABO+Rh type: A+, A−, B+, B−, AB+, AB−, O+, O−
Contact_Number	Integer	No	10-digit contact number
Email	String	No	Donor email address
City	String	No	City of residence (20 cities)
State	String	No	State of residence (15 states)
Country	String	No	Country — all values: India
Last_Donation_Date	Date/String	Yes	Date of last donation; 'Never' if donor has not yet donated
Total_Donations	Integer	No	Cumulative count of donations (0–9)
Eligible_for_Donation	Categorical	No	Yes / No — current eligibility status
Medical_Condition	String	Yes	Disqualifying medical condition if any; blank if none
Weight_kg	Float	No	Donor body weight in kilograms (45–110 kg)
Hemoglobin_g_dL	Float	No	Hemoglobin level in g/dL (10.0–18.0)
Donation_Center	String	No	Name of the registered donation centre (10 centres)
Registration_Date	Date	No	Date the donor registered in the system (2008–2025)

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5. Key Performance Indicators (KPIs)

The following high-level KPIs are surfaced on the dashboard:

10,000 Total Donors	**6,416** Eligible Donors	**2.08** Avg. Donations	**15** States Covered
O+ (37.3%) Most Common Group	**Maharashtra** Top State	**AB− (1.0%)** Rarest Group	**10** Donation Centers

6. Demographics Analysis

6.1 Blood Group Distribution

Blood group distribution follows patterns typical of the Indian population, with O+ and B+ collectively representing over 68% of all donors. Rare negative groups remain significantly underrepresented.

Blood Group	Count	% of Total	Category
O+	3,729	37.3%	Most common — universal compatible recipient
B+	3,136	31.4%	Second most common
A+	1,842	18.4%	Third most common
O−	419	4.2%	Universal donor — high strategic value
AB+	371	3.7%	Universal recipient
A−	208	2.1%	Rare negative group
B−	194	1.9%	Rare negative group
AB−	101	1.0%	Rarest group in dataset — critical shortage risk
Grand Total	10,000	100%	—

Key insight: Negative blood groups together represent only 11.2% of donors (1,122 donors). Since negative blood can only be transfused from negative donors in many cases, targeted recruitment campaigns for Rh-negative donors — especially AB− and B− — are strongly recommended.

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6.2 Gender Distribution

Gender is broadly balanced across the dataset, indicating inclusive outreach. Female donors marginally outnumber males.

Gender	Count	% of Total	Notes
Female	4,986	49.9%	Slightly larger group
Male	4,906	49.1%	Near parity with female
Other	108	1.1%	Non-binary / gender-diverse donors
Total	10,000	100%	—

6.3 Age Distribution

Donors range from 18 to 65 years with a mean age of 33.9 years. The 26–38 age cohort dominates, reflecting strong participation from working-age adults.

Age Band	Count	% of Total	Insight
18–24	1,779	17.8%	Young first-time donors
25–31	2,357	23.6%	High engagement group
32–38	2,711	27.1%	Largest single cohort
39–45	2,000	20.0%	Strong mid-career segment
46–52	854	8.5%	Declining participation
53–59	257	2.6%	Lower activity
60–66	42	0.4%	Minimal but present

6.4 Medical Conditions & Eligibility

35.8% of donors (3,584) are currently ineligible for donation. Medical conditions are the primary driver of ineligibility. Of 10,000 donors, 2,195 have a recorded medical condition.

Medical Condition	Count	% of Donors	Eligibility Impact
None (Healthy)	7,805	78.1%	Generally eligible
Hypertension	614	6.1%	Disqualifying

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Medical Condition	Count	% of Donors	Eligibility Impact
Anemia	469	4.7%	Disqualifying (Hb < threshold)
Diabetes	388	3.9%	Disqualifying
Asthma	308	3.1%	Conditionally disqualifying
Recent Surgery	207	2.1%	Temporary deferral
Infection (resolved)	109	1.1%	Temporary deferral
Cardiac Issues	100	1.0%	Disqualifying

7. Geographic Analysis

7.1 State-Level Distribution

Donors are spread across 15 Indian states. Maharashtra and Gujarat dominate, together accounting for 29.1% of the total donor pool, reflecting large urban populations in Mumbai, Pune, Ahmedabad, and Surat.

State	Count	% Share	Key Cities
Maharashtra	1,460	14.6%	Mumbai, Pune, Nagpur
Gujarat	1,452	14.5%	Ahmedabad, Surat, Vadodara
Uttar Pradesh	1,003	10.0%	Kanpur, Lucknow
West Bengal	537	5.4%	Kolkata
Punjab	533	5.3%	Mohali
Bihar	519	5.2%	Patna
Rajasthan	517	5.2%	Jaipur
Delhi	511	5.1%	Delhi
Karnataka	507	5.1%	Bengaluru
Haryana	504	5.0%	Gurgaon
Tamil Nadu	500	5.0%	Chennai
Andhra Pradesh	495	5.0%	Visakhapatnam
Kerala	490	4.9%	Thiruvananthapuram
Telangana	488	4.9%	Hyderabad

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State	Count	% Share	Key Cities
Madhya Pradesh	484	4.8%	Bhopal
Grand Total	10,000	100%	—

7.2 City-Level Average Donations

Average donations per city range from 1.96 (Gurgaon) to 2.20 (Patna), revealing meaningful variation in donor engagement. Cities with higher averages suggest a more loyal, repeat-donation culture.

City	Donors	Avg. Donations	Relative Performance
Patna	519	2.20	Highest engagement — top city
Hyderabad	488	2.20	Tied highest engagement
Kolkata	537	2.18	Strong repeat donors
Thiruvananthapuram	490	2.18	High loyalty
Jaipur	517	2.16	Above average
Gurgaon	504	1.96	Lowest — retention risk
Mumbai	495	2.00	Below average
Bhopal	484	2.00	Below average

8. Donation Centre Performance

Ten donation centres are represented, each with similar donor volumes (951–1,039), indicating fairly uniform distribution — possibly reflecting quota-based or stratified sampling in the dataset. However, average donations per donor vary meaningfully, revealing differences in donor retention.

Donation Centre	Donor Count	Avg. Donations	Ranking
Corporate Health Centre	1,039	2.18	🥇 Highest retention
NABL Blood Centre	1,032	2.13	2nd
Community Blood Bank	951	2.10	3rd
Red Cross Blood Bank	1,033	2.10	4th
Fortis Blood Bank	971	2.10	5th
Apollo Hospitals Blood Bank	973	2.08	6th — at average
Private Diagnostic Centre	1,030	2.04	7th

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Donation Centre	Donor Count	Avg. Donations	Ranking
Government Blood Bank	968	2.05	8th
City Blood Centre	1,003	2.03	9th
Metro Blood Bank	1,000	2.03	Lowest retention
Grand Total / Average	10,000	2.08	Baseline

Observation: Corporate Health Centre leads in both donor volume and average donations, suggesting strong follow-up processes or a more engaged donor community. Metro Blood Bank and City Blood Centre trail in average donations and would benefit from targeted re-engagement strategies.

9. Physical & Clinical Metrics

9.1 Body Weight

Metric	Value	Metric	Value
Minimum	45.0 kg	Mean	68.0 kg
25th Percentile	61.1 kg	Median	67.9 kg
75th Percentile	74.9 kg	Maximum	110.0 kg

All donors meet the minimum weight threshold of 45 kg required for safe blood donation.

9.2 Hemoglobin Levels

Metric	Value	Metric	Value
Minimum	10.0 g/dL	Mean	13.7 g/dL
25th Percentile	12.8 g/dL	Median	13.7 g/dL
75th Percentile	14.6 g/dL	Maximum	18.0 g/dL

The WHO minimum hemoglobin for blood donation is 12.5 g/dL (female) and 13.0 g/dL (male). The dataset mean of 13.7 g/dL is above threshold, though donors recording 10.0–12.4 g/dL may explain part of the ineligibility pool.

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10. Donation Frequency Analysis

Total donations per donor range from 0 to 9, with a mean of 2.08 and median of 2.

Segment	Count	% of Total	Description
0 Donations	576	5.8%	Registered but never donated
1 Donation	3,448	34.5%	First-time donors only
2–4 Donations	~4,800	~48%	Regular donors
5+ Donations	~1,176	~11.8%	Champion / loyal donors
Maximum	9 Donations	—	Most committed donors

576 donors (5.8%) have zero recorded donations — these represent newly registered or deferred donors and may warrant proactive outreach. The largest segment at 34.5% are first-time donors, indicating significant room to convert one-time donors into repeat contributors.

11. Pivot Table Layer Summary

The 'Table1 (2)' sheet contains multiple embedded pivot tables that serve as the data engine for all charts in the Dashboard, Demographics, and Geographics sheets. These are listed below:

Pivot Table	Metric	Purpose
Blood Group Count	Count of Blood_Group	Drives blood group distribution chart
Gender Count	Count of Gender	Drives gender distribution chart
State Count (x2 copies)	Count of State	Drives geographic map/bar chart
Donation Centre Volume	Count of Total_Donations	Centre comparison bar chart
Eligibility Split	Count (Yes/No)	Eligibility donut/bar chart
Age Band (All Donors)	Count of Total_Donations	Age demographic histogram
Age Band (Other Gender)	Count of Total_Donations	Filtered age band for 'Other' gender
Gender (duplicate)	Count of Gender	Supporting slicer interactions
Centre Avg. Donations	Average of Total_Donations	Centre retention KPI
Centre Count (repeat)	Count of Total_Donations	Cross-check / slicer target
City Avg & Count	Count + Average	City performance table in Geographics

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12. Recommendations & Insights

12.1 Blood Supply Strategy

- Launch a targeted recruitment drive for O−, A−, and B− donors — these three groups together represent only 4% of donors despite their critical role in emergency transfusions.
- Prioritise AB− recruitment: at just 101 donors (1.0%), this is the most under-represented group.
- Consider dedicated campaigns in Gujarat and Maharashtra for rare blood types given their large donor bases.

12.2 Donor Retention

- 576 donors with zero donations need targeted follow-up — email or SMS outreach could convert a significant portion.
- 3,448 first-time donors (34.5%) represent the single largest conversion opportunity; a 'second donation' reminder programme could materially increase average donations.
- Metro Blood Bank and City Blood Centre have the lowest average donations per donor (2.03) and should review their post-donation engagement processes.

12.3 Medical Condition Management

- Hypertension (614 donors) is the leading disqualifying condition. Partner with health organisations to offer blood pressure management programmes to convert deferred donors back to active status.
- Anemia accounts for 469 deferrals. Iron supplementation guidance and pre-screening could reduce this rate.
- 207 donors deferred for 'Recent Surgery' are likely to return — implement automated eligibility-reinstatement alerts at the 6-month mark.

12.4 Geographic Expansion

- Patna and Hyderabad show the highest donor engagement (avg. 2.20 donations each) — study and replicate these centres' practices in lower-performing cities.
- Gurgaon records the lowest average (1.96) despite a substantial donor count of 504 — an engagement programme is warranted.
- States like Telangana (488) and Madhya Pradesh (484) have the smallest representation and may benefit from new centre openings or mobile donation camps.

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13. Data Quality Notes

- Last_Donation_Date contains the string 'Never' for 576 donors with zero donations. This requires special handling in date-based calculations; these rows should be treated as NULL dates.
- Medical_Condition is blank for 7,805 records (78.1%) — this is by design and represents healthy donors, not missing data.
- The dataset spans 2008–2025 and is synthetically generated for analytical purposes; recency bias should be considered when interpreting registration trends.
- Gender category 'Other' (108 donors, 1.1%) is correctly represented but appears in a separate age-band pivot, suggesting the dashboard supports gender-filtered demographic views.
- Duplicate pivot tables for State and Donation Centre appear in Table1 (2) — these likely serve different slicer configurations on dashboard sheets.
- No null values detected in required fields (Donor_ID, Gender, Age, Blood_Group, Weight, Hemoglobin).

14. Technical Notes

14.1 Dashboard Architecture

The workbook follows a standard layered analytics architecture:

- Layer 1 — Raw Data ('Data' sheet): The single source of truth. All records reside here. No calculations are performed in this sheet.
- Layer 2 — Pivot Engine ('Table1 (2)' sheet): Twelve pivot tables aggregate the raw data into summary statistics consumed by charts.
- Layer 3 — Visual Dashboards ('Dashboard', 'Demographics', 'Geographics' sheets): Charts, KPI cards, and slicers that render the pivot data. These sheets appear to use Excel Slicers (detected via openpyxl warnings) for interactivity.
- Layer 4 — Metadata ('About' sheet): Describes the project context and instructions.

14.2 Interactivity

Excel Slicer List extensions are present in the workbook (visible in the file structure), indicating the dashboard uses slicers for filtering. Likely slicer fields include: Blood_Group, Gender, State, Donation_Center, and Eligible_for_Donation. These allow dashboard consumers to dynamically filter all connected charts simultaneously.

14.3 Data Volume & Performance

At 10,000 rows × 18 columns (180,000 cells), the dataset is lightweight and should perform well even in Excel without any optimisation. Calculated fields and pivot caches should remain fast. If the dataset is expanded to 100,000+ rows in future, consider migrating pivot sources to Power Query or Power Pivot for improved performance.

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15. Glossary

Term	Definition
ABO System	Blood classification system using A, B, AB, O antigens
Rh Factor	Presence (+) or absence (−) of the Rhesus D antigen on red blood cells
Universal Donor	O− blood type; can be transfused to any recipient in emergencies
Hemoglobin (Hb)	Protein in red blood cells that carries oxygen; measured in g/dL
Donor Deferral	Temporary or permanent exclusion from donation due to medical or lifestyle factors
KPI	Key Performance Indicator — a quantifiable measure of performance
Pivot Table	Excel aggregation tool that summarises data without altering the source
Slicer	Interactive Excel filter control that applies to connected pivot tables/charts
NABL	National Accreditation Board for Testing and Calibration Laboratories (India)

16. Document Control

Version	Date	Author	Changes
1.0	April 2025	Chevalla Nithin	Initial release

This document is generated from automated analysis of the source Excel file 'Blood_Donation_Analytics_Project_.xlsx'. All figures are derived directly from the dataset and are accurate as of the analysis date.

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